

On the ratio of surjective group homomorphisms to ring homomorphisms between cyclic rings

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Abstract: For $n \mid m$ there are $\varphi(n)$ surjective group homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ and $2^{\omega(n)}$ ring homomorphisms. We show that their ratio $\rho(n)$ is $2^{\varepsilon(n)}$ times the number of squares in $(\mathbb{Z}/n\mathbb{Z})^\times$, where $\varepsilon(n) \in \{-1, 0, 1\}$ is fixed by the power of 2 dividing n , and that for odd n the ring homomorphisms correspond to the subfields of degree at most 2 of $\mathbb{Q}(\zeta_n)$. We give two-term asymptotic formulas, with explicit constants, for the number of $n \leq x$ with $\rho(n) \notin \mathbb{Z}$ and for $\sum_{n \leq x} \rho(n)$, and check them against exact computations up to $2 \cdot 10^{10}$ and 10^{10} , respectively.

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1 Introduction

Let $n \mid m$. Gallian and Van Buskirk [1] showed that there are exactly $2^{\omega(n)}$ ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$, where $\omega(n)$ is the number of distinct prime factors of n , and it is elementary that exactly $\varphi(n)$ of the group homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ are surjective. This paper is about the ratio

$$\rho(n) = \frac{\varphi(n)}{2^{\omega(n)}}. \quad (1)$$

It is often an integer ($\rho(35) = 6$, $\rho(120) = 4$) and sometimes not ($\rho(6) = \frac{1}{2}$, $\rho(98) = \frac{21}{2}$). We determine $\rho(n)$ exactly as a group-theoretic quantity, describe the integers where it fails to be integral, and give two-term asymptotic formulas for how often this happens and for the mean value of ρ .

Write $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$ and $U_n^2 = \{u^2 : u \in U_n\}$, so that $|U_n^2|$ is the number of quadratic residues modulo n that are prime to n . For $n = 2^s n'$ with n' odd put

$$\varepsilon(n) = \begin{cases} 0, & s = 0 \text{ or } s = 2, \\ -1, & s = 1, \\ 1, & s \geq 3, \end{cases} \quad (2)$$

and let

$$\mathcal{E} = \{n \equiv 2 \pmod{4} : \text{every prime divisor of } n/2 \text{ is } \equiv 3 \pmod{4}\}, \quad (3)$$

so that $\mathcal{E} = \{2, 6, 14, 18, 22, 38, 42, 46, 54, \dots\}$.

Theorem 1.1. *For every $n \geq 1$,*

$$\rho(n) = 2^{\varepsilon(n)} |U_n^2|.$$

Moreover, $\rho(n)$ is an integer if and only if $n \notin \mathcal{E}$, and $\rho(n) = |U_n^2|$ whenever $n \not\equiv 2 \pmod{4}$ and $8 \nmid n$.

For odd n the identity $|U_n^2| = \varphi(n)/2^{\omega(n)}$ is classical, and Theorem 1.1 is a short consequence of the structure of U_n . We record it because it gives the exact form of $\rho(n)$ for every n , and the factor $2^{\varepsilon(n)}$ is precisely what produces the set $\mathcal{E}$. Our second result explains the count $2^{\omega(n)}$ on the field side. Let $\zeta_n = e^{2\pi i/n}$ and, for an odd prime p , let $p^* = (-1)^{(p-1)/2}p$.

Theorem 1.2. *Let n be odd and $n \mid m$. For a ring homomorphism $f: \mathbb{Z}_m \rightarrow \mathbb{Z}_n$ let $S(f)$ be the set of primes $p \mid n$ with $f(1) \equiv 1 \pmod{p}$, and put $d_f = \prod_{p \in S(f)} p^*$. Then $f \mapsto \mathbb{Q}(\sqrt{d_f})$ is a bijection from the set of ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ onto the set of subfields $K \subseteq \mathbb{Q}(\zeta_n)$ with $[K : \mathbb{Q}] \leq 2$. Moreover $\rho(n) = [\mathbb{Q}(\zeta_n) : M_n]$, where M_n is the compositum of these fields.*

The bijection comes from an isomorphism between the Boolean group of idempotents of $\mathbb{Z}/n\mathbb{Z}$ and the group of quadratic characters of U_n (Proposition 4.3). Figure 1 shows it for $n = 105$. For even n the two groups have orders $2^{\omega(n)}$ and $2^{\omega(n)+\varepsilon(n)}$, which is another way of reading Theorem 1.1.

The remaining results are analytic. Let $\mathcal{E}(x) = \#\{n \leq x : n \in \mathcal{E}\}$, let χ_{-4} be the non-principal character modulo 4, and let γ be Euler's constant.

Theorem 1.3. *As $x \rightarrow \infty$,*

$$\mathcal{E}(x) = \frac{Cx}{\sqrt{\log x}} \left(1 + \frac{\kappa}{\log x} + O\left(\frac{1}{(\log x)^2}\right) \right),$$

where

$$C = \frac{1}{\pi\sqrt{2}} \prod_{p \equiv 3(4)} \left(1 - \frac{1}{p^2}\right)^{-1/2} = 0.24325\,99441\,92945\dots,$$

$$\kappa = \frac{1}{2} - \frac{\gamma}{4} + \frac{\log 2}{4} + \frac{1}{2} \sum_{p \equiv 3(4)} \frac{\log p}{p^2 - 1} + \frac{1}{4} \frac{L'(1, \chi_{-4})}{L(1, \chi_{-4})} = 0.70475\,34517\,05947\dots$$

Theorem 1.4. *As $x \rightarrow \infty$,*

$$\sum_{n \leq x} \rho(n) = \frac{C_2 x^2}{\sqrt{\log x}} \left(1 + \frac{\kappa_2}{\log x} + O\left(\frac{1}{(\log x)^2}\right) \right),$$

where

$$C_2 = \frac{1}{2\sqrt{\pi}} \prod_p \left(1 + \frac{1}{2p}\right) \left(1 - \frac{1}{p}\right)^{1/2} = 0.22909\,07914\,88938\dots,$$

$$\kappa_2 = \frac{1}{4} - \frac{\gamma}{4} - \frac{1}{4} \sum_p \frac{\log p}{(2p+1)(p-1)} = 0.02482\,64240\,33851\dots$$

Both main terms fit into known results. The set $\mathcal{E}$ is twice the set of odd integers composed of primes $\equiv 3 \pmod{4}$. Counting problems of this kind go back to Landau's work on sums of two squares, and such counting functions have asymptotic expansions in powers of $1/\log x$ (see Moree [2]); the leading constant $2C$ for our set appears explicitly in the work of Jones and Zvonkin [3]. Finch and Sebah [4] obtained the leading term of $\sum_{n \leq x} |U_n^2|$; since $\rho(n)$ and $|U_n^2|$ differ only by the factor $2^{\varepsilon(n)}$, their constant is $\frac{43}{40}C_2$. What we add is closed forms for the second-order coefficients κ and κ_2 , self-contained proofs of both two-term expansions by the Selberg–Delange method [5, Theorem II.5.2], and a comparison with exact computations of $\mathcal{E}(x)$ up to $x = 2 \cdot 10^{10}$ and of $\sum_{n \leq x} \rho(n)$ up to $x = 10^{10}$.

Section 2 collects the counting lemmas. Theorem 1.1 is proved in Section 3, Theorem 1.2 in Section 4, and Theorems 1.3 and 1.4 in Section 5, which ends with the numerical comparison.

2 Preliminaries

Ring homomorphisms are not required to send 1 to 1; this is the convention of [1].¹ We write $B_n = \{e \in \mathbb{Z}/n\mathbb{Z} : e^2 = e\}$ for the set of idempotents.

Lemma 2.1. *Let $n \mid m$. The map $f \mapsto f(1)$ is a bijection from the ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ onto B_n , and $|B_n| = 2^{\omega(n)}$.*

Proof. An additive map satisfies $f(k) = kf(1)$, and multiplicativity gives $f(1) = f(1)^2$. Conversely, if $e \in B_n$ then $k \mapsto ke$ is well defined on $\mathbb{Z}_m$ because $n \mid m$, it is additive, and $(ab)e = (ab)e^2 = (ae)(be)$. Write $n = \prod_{i=1}^k p_i^{r_i}$. By the Chinese remainder theorem $\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{r_i}\mathbb{Z}$, and each factor is a local ring, whose only idempotents are 0 and 1. Hence $|B_n| = 2^k$. $\square$

The proof shows more. The map $e \mapsto S(e) = \{p_i : e \equiv 1 \pmod{p_i}\}$ is a bijection from B_n onto the subsets of $\{p_1, \dots, p_k\}$; note that $e \equiv 1 \pmod{p_i}$ forces $e \equiv 1 \pmod{p_i^{r_i}}$, since $e \pmod{p_i^{r_i}} \in \{0, 1\}$. With the Boolean sum $e \oplus e' = e + e' - 2ee'$, which reduces to addition modulo 2 in each component, B_n is an elementary abelian 2-group and

$$S(e \oplus e') = S(e) \triangle S(e'). \quad (4)$$

Lemma 2.2. *Let $n \mid m$. Exactly $\varphi(n)$ group homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ are surjective.*

Proof. Group homomorphisms are $k \mapsto ka$ with $a \in \mathbb{Z}_n$ (well defined since $n \mid m$), and $k \mapsto ka$ is onto exactly when $\gcd(a, n) = 1$. $\square$

Lemma 2.3. *For every $n \geq 1$, $[U_n : U_n^2] = 2^{\omega(n) + \varepsilon(n)}$.*

Proof. For a finite abelian group A the squaring map has kernel $A[2]$, so $[A : A^2] = |A[2]|$, and a cyclic group contributes a factor 2 to $|A[2]|$ if its order is even and 1 otherwise. By the Chinese remainder theorem $U_n \cong \prod_{p^r \parallel n} U_{p^r}$. For odd p the group U_{p^r} is cyclic of even order $p^{r-1}(p-1)$; moreover $U_2 = 1$, $U_4 \cong \mathbb{Z}/2$, and $U_{2^s} \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{s-2}$ for $s \geq 3$ [6, Chapter 4]. If $n = 2^s n'$ with n' odd, this gives $[U_n : U_n^2] = 2^{\omega(n') + t}$ with $t = 0, 0, 1, 2$ for $s = 0, s = 1, s = 2, s \geq 3$. Since $\omega(n) = \omega(n') + [s \geq 1]$, the exponent is $\omega(n) + \varepsilon(n)$ in every case. $\square$

3 The exact value of $\rho(n)$

Proof of Theorem 1.1. By Lemma 2.3, $\varphi(n) = |U_n| = |U_n^2| [U_n : U_n^2] = |U_n^2| 2^{\omega(n) + \varepsilon(n)}$. Dividing by $2^{\omega(n)}$ gives $\rho(n) = 2^{\varepsilon(n)} |U_n^2|$.

If $\varepsilon(n) \geq 0$, then $\rho(n)$ is an integer and $n \notin \mathcal{E}$. Let $\varepsilon(n) = -1$, that is $n = 2n'$ with n' odd. Then $U_n \cong U_{n'}$, so $\rho(n) = \frac{1}{2} |U_{n'}^2|$ with

$$|U_{n'}^2| = \prod_{p^r \parallel n'} \frac{p^{r-1}(p-1)}{2}$$

by Lemma 2.3 applied to each cyclic factor. A factor $p^{r-1}(p-1)/2$ is even exactly when $p \equiv 1 \pmod{4}$. So $\rho(n) \in \mathbb{Z}$ if and only if some prime divisor of n' is $\equiv 1 \pmod{4}$, which means $n \notin \mathcal{E}$. For $n = 2$ the product is empty, so $\rho(2) = \frac{1}{2}$, and indeed $2 \in \mathcal{E}$. The last assertion is the case $\varepsilon(n) = 0$. $\square$

Equivalently, $2^{\omega(n)} \mid \varphi(n)$ if and only if $n \notin \mathcal{E}$. Table 1 shows each case. Note that $\rho(n)$ and $|U_n^2|$ really do differ when $n \equiv 2 \pmod{4}$ or $8 \mid n$: for $n = 26$ the ratio is 3 while there are 6 unit squares, and for $n = 120$ the ratio is 4 while there are 2.

¹If one insists on $f(1) = 1$, there is exactly one ring homomorphism $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ when $n \mid m$, and the question studied here disappears.

Table 1. The quantities of Theorem 1.1 for some small n .

n	factorisation	$\varepsilon(n)$	$\varphi(n)$	$2^{\omega(n)}$	$ U_n^2 $	$\rho(n)$	$n \in \mathcal{E}$
2	2	-1	1	2	1	1/2	yes
6	$2 \cdot 3$	-1	2	4	1	1/2	yes
10	$2 \cdot 5$	-1	4	4	2	1	no
12	$2^2 \cdot 3$	0	4	4	1	1	no
24	$2^3 \cdot 3$	1	8	4	1	2	no
26	$2 \cdot 13$	-1	12	4	6	3	no
35	$5 \cdot 7$	0	24	4	6	6	no
45	$3^2 \cdot 5$	0	24	4	6	6	no
98	$2 \cdot 7^2$	-1	42	4	21	21/2	yes
105	$3 \cdot 5 \cdot 7$	0	48	8	6	6	no
120	$2^3 \cdot 3 \cdot 5$	1	32	8	2	4	no

4 The cyclotomic picture

Let $G_n = \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$. The map $u \mapsto \sigma_u$, $\sigma_u(\zeta_n) = \zeta_n^u$, is an isomorphism $U_n \rightarrow G_n$ [7, Chapter 2], and we identify the two groups. Let $X_n = \text{Hom}(U_n, \{\pm 1\})$ be the group of characters of order at most 2. Since U_n/U_n^2 is an elementary abelian 2-group, $|X_n| = [U_n : U_n^2]$, and $\chi \mapsto \ker \chi$ is a bijection from X_n onto the subgroups of index at most 2 in U_n .

Definition 4.1. M_n is the fixed field of U_n^2 in $\mathbb{Q}(\zeta_n)$.

Proposition 4.2. Let $n \geq 1$ and put $w = \omega(n) + \varepsilon(n)$.

- (i) A subfield $K \subseteq \mathbb{Q}(\zeta_n)$ satisfies $K \subseteq M_n$ if and only if $\text{Gal}(K/\mathbb{Q})$ has exponent at most 2. In particular M_n is the compositum of all quadratic subfields of $\mathbb{Q}(\zeta_n)$.
- (ii) $[M_n : \mathbb{Q}] = 2^w$, $[\mathbb{Q}(\zeta_n) : M_n] = |U_n^2|$, and $\mathbb{Q}(\zeta_n)$ has exactly $2^w - 1$ quadratic subfields.
- (iii) $M_n = \mathbb{Q}(\{\sqrt{p^*} : p \mid n, p \text{ odd}\} \cup \{\sqrt{-1} : 4 \mid n\} \cup \{\sqrt{2} : 8 \mid n\})$.

Proof. (i) Every subfield K is Galois over $\mathbb{Q}$, and $\text{Gal}(K/\mathbb{Q}) \cong U_n/H$ where H fixes K . This quotient has exponent at most 2 if and only if $H \supseteq U_n^2$, that is $K \subseteq M_n$. In the elementary abelian 2-group U_n/U_n^2 the trivial subgroup is the intersection of the subgroups of index 2, so U_n^2 is the intersection of the subgroups of index 2 of U_n that contain it. By the Galois correspondence, M_n is the compositum of the quadratic subfields of $\mathbb{Q}(\zeta_n)$.

(ii) The degrees follow from Lemma 2.3 and the Galois correspondence. The quadratic subfields correspond to subgroups of index 2 containing U_n^2 , that is to the $2^w - 1$ non-trivial elements of X_n .

(iii) Let $a_1, \dots, a_t$ be the listed radicands. Each $\sqrt{a_j}$ lies in $\mathbb{Q}(\zeta_n)$: for odd $p \mid n$ the Gauss sum $g_p = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^a$ satisfies $g_p^2 = p^*$ [6, Proposition 6.3.2]; $\sqrt{-1} = \zeta_4$; and $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$. Counting the radicands gives $t = \omega(n') + [4 \mid n] + [8 \mid n] = w$, where n' is the odd part of n . The $2^w - 1$ numbers $\prod_{j \in J} a_j$ with $\emptyset \neq J \subseteq \{1, \dots, t\}$ are pairwise distinct modulo $\mathbb{Q}^{\times 2}$. Indeed, a quotient of two of them is, up to a square, again a product over a non-empty set of radicands, and such a product is never a square: an odd prime in it occurs to the first power, and without odd primes it is -1 , 2 or -2 . So they define $2^w - 1$ distinct quadratic subfields of $\mathbb{Q}(\zeta_n)$, which by (ii) are all of them. By (i) their compositum is M_n , and it equals $\mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_t})$. $\square$

We call M_n the maximal multiquadratic subfield of $\mathbb{Q}(\zeta_n)$.² For odd p and $u \in U_n$,

$$\sigma_u(g_p) = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^{ua} = \left(\frac{u}{p}\right) \sum_{a=1}^{p-1} \left(\frac{ua}{p}\right) \zeta_p^{ua} = \left(\frac{u}{p}\right) g_p. \quad (5)$$

²For odd n with k distinct prime factors, $[M_n : \mathbb{Q}] = 2^k$, so M_n is biquadratic only when $k = 2$.

From now on let n be odd. For $e \in B_n$ define

$$\chi_e(u) = \prod_{p \in S(e)} \left(\frac{u}{p}\right) \quad (u \in U_n), \quad d_e = \prod_{p \in S(e)} p^*, \quad \gamma_e = \prod_{p \in S(e)} g_p,$$

so that $\chi_0 = 1$, $d_0 = 1$, $\gamma_0 = 1$.

Proposition 4.3. *Let n be odd. Then $e \mapsto \chi_e$ is a group isomorphism $(B_n, \oplus) \rightarrow X_n$, and for every $e \in B_n$ the fixed field of $\ker \chi_e$ is $\mathbb{Q}(\sqrt{d_e})$.*

Proof. Each χ_e is a product of characters of order dividing 2, so $\chi_e \in X_n$. By (4), $\chi_{e \oplus e'} = \chi_e \chi_{e'}$, because a prime in $S(e) \cap S(e')$ contributes $(\frac{u}{p})^2 = 1$. If $e \neq 0$, choose $p \in S(e)$, a quadratic non-residue v modulo p , and $u \in U_n$ with $u \equiv v \pmod{p}$ and $u \equiv 1$ modulo every other prime power dividing n ; then $\chi_e(u) = -1$. Hence the homomorphism is injective. Both groups have order $2^{\omega(n)}$, by Lemma 2.1 and by $|X_n| = [U_n : U_n^2]$ with Lemma 2.3, so it is bijective.

By (5), $\sigma_u(\gamma_e) = \chi_e(u)\gamma_e$, and $\gamma_e^2 = d_e$. Hence σ_u fixes $\mathbb{Q}(\gamma_e) = \mathbb{Q}(\sqrt{d_e})$ exactly when $u \in \ker \chi_e$, and the Galois correspondence gives the claim. $\square$

Proof of Theorem 1.2. Compose three bijections: $f \mapsto f(1)$ from ring homomorphisms onto B_n (Lemma 2.1); $e \mapsto \ker \chi_e$ from B_n onto the subgroups of index at most 2 in U_n (Proposition 4.3); and the Galois correspondence from these subgroups onto the subfields of degree at most 2. By Proposition 4.3 the composite sends f to $\mathbb{Q}(\sqrt{d_{f(1)}}) = \mathbb{Q}(\sqrt{d_f})$. Finally, $\varepsilon(n) = 0$ for odd n , so Theorem 1.1 and Proposition 4.2(ii) give $\rho(n) = |U_n^2| = [\mathbb{Q}(\zeta_n) : M_n]$. $\square$

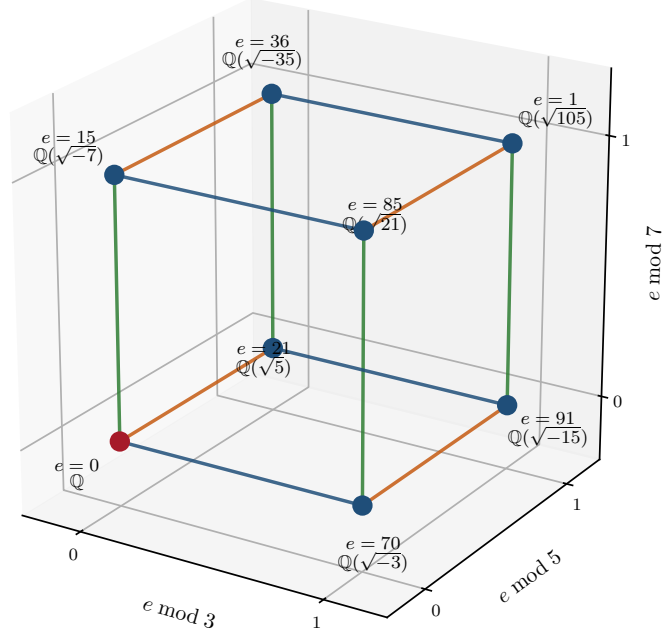


Figure 1. Theorem 1.2 for $n = 105$. The eight idempotents e of $\mathbb{Z}/105\mathbb{Z}$ sit at the vertices ($e \bmod 3$, $e \bmod 5$, $e \bmod 7$) of the cube, and each is labelled with the field $\mathbb{Q}(\sqrt{d_e}) \subseteq \mathbb{Q}(\zeta_{105})$. Moving along an edge changes one coordinate, which is the Boolean sum with 70, 21 or 15 (blue, orange and green edges), and multiplies d_e by -3 , 5 or -7 modulo squares. The compositum of the seven quadratic fields is $M_{105} = \mathbb{Q}(\sqrt{-3}, \sqrt{5}, \sqrt{-7})$, and $[\mathbb{Q}(\zeta_{105}) : M_{105}] = 48/8 = 6 = \rho(105)$.

Remark 4.4. For even n the groups B_n and X_n have orders $2^{\omega(n)}$ and $2^{\omega(n)+\varepsilon(n)}$. When $4 \parallel n$ they are still isomorphic: the 2-component of e is matched with the character $u \mapsto (-1)^{(u-1)/2}$, whose field is $\mathbb{Q}(\sqrt{-1})$. When $n \equiv 2 \pmod{4}$ the 2-component of e has no character to go to, because U_2 is trivial, and when $8 \mid n$ there are two independent characters of U_{2^s} and only one idempotent component. This mismatch is the factor $2^{\varepsilon(n)}$ in Theorem 1.1.

Figure 2 shows the same structure inside a single unit group. With generators 2 of U_5 and 3 of U_7 , the group $U_{35} \cong \mathbb{Z}/4 \times \mathbb{Z}/6$ is drawn on a torus. The two characters $(\frac{\cdot}{5})$ and $(\frac{\cdot}{7})$ split it into four cosets of U_{35}^2 , and the coset of squares has $6 = \rho(35)$ elements.

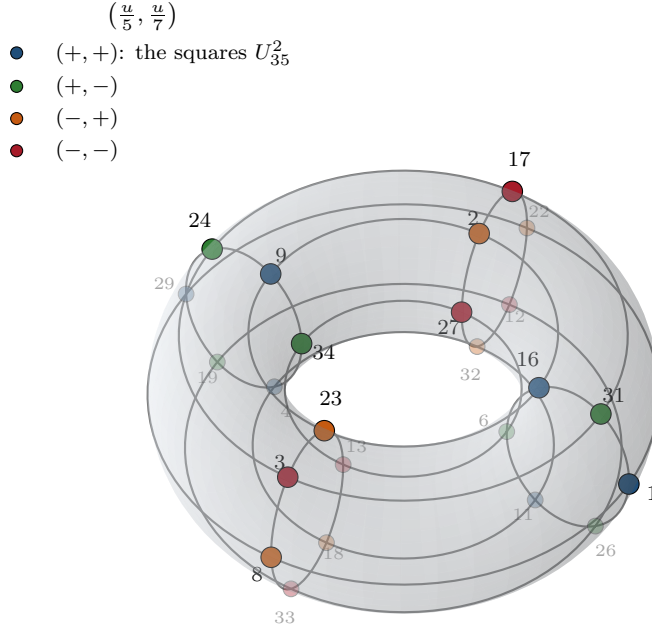


Figure 2. The 24 elements of U_{35} on a torus: u is placed at angle $2\pi k/4$ around the core circle and $2\pi l/6$ around the tube, where $u \equiv 2^k \pmod{5}$ and $u \equiv 3^l \pmod{7}$. The colour records $((\frac{u}{5}), (\frac{u}{7}))$. The six blue points $\{1, 4, 9, 11, 16, 29\}$ form U_{35}^2 , the subgroup fixing $M_{35} = \mathbb{Q}(\sqrt{5}, \sqrt{-7})$. Labels of points on the far side of the torus are drawn faint.

5 Distribution

Write $s = \sigma + i\tau$. We use the Selberg–Delange method in the following form [5, Theorem II.5.2], specialised to non-negative coefficients and two terms.

Lemma 5.1. *Let $a_n \geq 0$ and let $F(s) = \sum_{n \geq 1} a_n n^{-s}$ converge for $\sigma > 1$. Suppose there are $z > 0$, $c_0 > 0$, $0 < \delta \leq 1$ and $M > 0$ such that $G(s) = F(s)\zeta(s)^{-z}$ extends holomorphically to the region $D = \{\sigma \geq 1 - c_0/(1 + |\tau|)\}$ and satisfies $|G(s)| \leq M(1 + |\tau|)^{1-\delta}$ there. Put $Z(s) = ((s-1)\zeta(s))^z G(s)/s$. Then, for $x \geq 3$,*

$$\sum_{n \leq x} a_n = x(\log x)^{z-1} \left(\lambda_0 + \frac{\lambda_1}{\log x} + O\left(\frac{1}{(\log x)^2}\right) \right), \quad \lambda_0 = \frac{Z(1)}{\Gamma(z)}, \quad \lambda_1 = \frac{Z'(1)}{\Gamma(z-1)}.$$

For $z = \frac{1}{2}$ we have $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$. Since $(s-1)\zeta(s) = 1 + \gamma(s-1) + O((s-1)^2)$,

$$\frac{Z'(1)}{Z(1)} = \frac{\gamma}{2} + \frac{G'(1)}{G(1)} - 1, \quad \text{so} \quad \frac{\lambda_1}{\lambda_0} = -\frac{1}{2} \left(\frac{\gamma}{2} + \frac{G'(1)}{G(1)} - 1 \right) \quad \left(z = \frac{1}{2} \right). \quad (6)$$

5.1 The exceptional set

Let $\mathcal{A}$ be the set of odd $a \geq 1$ all of whose prime factors are $\equiv 3 \pmod{4}$. By (3), $\mathcal{E} = \{2a : a \in \mathcal{A}\}$, so $\mathcal{E}(x) = A(x/2)$ with $A(y) = \#\{a \leq y : a \in \mathcal{A}\}$. For $\sigma > 1$ put

$$P(s) = \sum_{a \in \mathcal{A}} a^{-s} = \prod_{p \equiv 3 \pmod{4}} (1 - p^{-s})^{-1}, \quad L(s) = L(s, \chi_{-4}).$$

Comparing Euler factors at $p = 2$, at $p \equiv 1 \pmod{4}$ and at $p \equiv 3 \pmod{4}$ gives, for $\sigma > 1$,

$$P(s)^2 = \frac{(1 - 2^{-s}) \zeta(s) P(2s)}{L(s)}. \quad (7)$$

Indeed, at $p \equiv 3 \pmod{4}$ the right side has factor $(1 - p^{-s})^{-1}(1 + p^{-s})(1 - p^{-2s})^{-1} = (1 - p^{-s})^{-2}$, and at the other primes all factors cancel. Iterating (7) at $s = 2, 4, 8, \dots$ gives a rapidly convergent formula,

$$P(2) = \prod_{j \geq 1} \left(\frac{(1 - 2^{-2^j}) \zeta(2^j)}{L(2^j)} \right)^{2^{-j}} = 1.16807\,55854\,10514\dots, \quad (8)$$

which is how the value of C was computed.³

Proof of Theorem 1.3. By (7), $G(s) = P(s)\zeta(s)^{-1/2}$ satisfies, for $\sigma > 1$,

$$G(s)^2 = R(s) := \frac{(1 - 2^{-s}) P(2s)}{L(s)}.$$

The product $P(2s)$ converges absolutely and has no zeros for $\sigma > \frac{1}{2}$, and $1 - 2^{-s} \neq 0$ for $\sigma > 0$. Since the modulus is fixed, there is $c > 0$ such that $L(s) \neq 0$ and $1/L(s) \ll \log(|\tau| + 3)$ for $\sigma \geq 1 - c/\log(|\tau| + 3)$ [8, Chapter 11]. Choose $c_0 > 0$ so small that the region D of Lemma 5.1 lies inside this zero-free region and inside $\sigma > \frac{3}{4}$. Then R is holomorphic and zero-free on D , and D is simply connected, so R has a holomorphic square root on D ; we take the one that agrees with G on $\sigma > 1$ (two continuous square roots of the same zero-free function on a connected set agree once they agree at one point). On D we have $|R(s)| \ll \log(|\tau| + 3)$, so $|G(s)| \ll (1 + |\tau|)^{1/2}$ and Lemma 5.1 applies with $z = \delta = \frac{1}{2}$.

Since $L(1) = \pi/4$ and $R(\sigma) > 0$ for real $\sigma \geq 1$,

$$Z(1) = G(1) = \sqrt{R(1)} = \left(\frac{2P(2)}{\pi} \right)^{1/2}, \quad \frac{G'(1)}{G(1)} = \frac{1}{2} \frac{R'(1)}{R(1)} = \frac{1}{2} \left(\log 2 - 2\Sigma_3 - \frac{L'(1)}{L(1)} \right),$$

where $\Sigma_3 = \sum_{p \equiv 3 \pmod{4}} \log p / (p^2 - 1) = -P'(2)/P(2)$. Lemma 5.1 gives

$$A(y) = \frac{y}{\sqrt{\log y}} \left(\lambda_0 + \frac{\lambda_1}{\log y} + O\left(\frac{1}{(\log y)^2} \right) \right), \quad \lambda_0 = \frac{\sqrt{2P(2)}}{\pi},$$

with λ_1/λ_0 given by (6). Put $y = x/2$. Since $(\log y)^{-1/2} = (\log x)^{-1/2} (1 + \frac{\log 2}{2 \log x} + O((\log x)^{-2}))$ and $(\log y)^{-1} = (\log x)^{-1} + O((\log x)^{-2})$,

$$\mathcal{E}(x) = \frac{\lambda_0}{2} \frac{x}{\sqrt{\log x}} \left(1 + \frac{\lambda_1/\lambda_0 + \frac{1}{2} \log 2}{\log x} + O\left(\frac{1}{(\log x)^2} \right) \right).$$

³The first factor involves $L(2)$, which is Catalan's constant. A direct product over the primes $p < 10^7$ agrees with (8) to eight decimal places, as the size of the omitted tail predicts.

Here $\lambda_0/2 = \sqrt{P(2)}/(\pi\sqrt{2}) = C$, and by (6)

$$\frac{\lambda_1}{\lambda_0} + \frac{\log 2}{2} = \frac{1}{2} - \frac{\gamma}{4} - \frac{1}{4} \left(\log 2 - 2\Sigma_3 - \frac{L'(1)}{L(1)} \right) + \frac{\log 2}{2} = \kappa. \quad \square$$

For the numerical value of κ we used $\Sigma_3 = 0.22873\,63531\,94032\dots$. Iterating (7) gives

$$\log P(s) = \sum_{j \geq 0} 2^{-j-1} \log \tilde{R}(2^j s), \quad \tilde{R}(s) = \frac{(1 - 2^{-s}) \zeta(s)}{L(s)},$$

hence the rapidly convergent series $\Sigma_3 = -\frac{1}{2} \sum_{j \geq 1} \tilde{R}'(2^j)/\tilde{R}(2^j)$. We also used

$$\frac{L'(1, \chi_{-4})}{L(1, \chi_{-4})} = \gamma + 2 \log 2 + 3 \log \pi - 4 \log \Gamma\left(\frac{1}{4}\right) = 0.24560\,95847\,77314\dots,$$

which follows from Lerch's formula for $L'(0, \chi_{-4})$ and the functional equation.⁴

5.2 The mean value of ρ

Put $g(n) = \rho(n)/n$. Since $\varphi(p^k)/p^k = 1 - 1/p$, the function g is multiplicative with $g(p^k) = \frac{1}{2}(1 - 1/p)$ for every prime p and $k \geq 1$, including $p = 2$. Its Dirichlet series is, for $\sigma > 1$,

$$F(s) = \sum_{n \geq 1} \frac{g(n)}{n^s} = \prod_p \left(1 + \frac{1 - p^{-1}}{2} \frac{p^{-s}}{1 - p^{-s}} \right) = \prod_p \frac{1 - \frac{1+p^{-1}}{2} p^{-s}}{1 - p^{-s}}. \quad (9)$$

Lemma 5.2. *Let $b > 0$ and $I_b(x) = \int_3^x t(\log t)^{-b} dt$. Then $I_b(x) \ll_b x^2(\log x)^{-b}$ and*

$$I_b(x) = \frac{x^2}{2(\log x)^b} \left(1 + \frac{b}{2 \log x} + O_b\left(\frac{1}{(\log x)^2}\right) \right).$$

Proof. Split at $\sqrt{x}$: the part over $[3, \sqrt{x}]$ is at most $x/2$, since $\log t \geq \log 3 > 1$, and the part over $[\sqrt{x}, x]$ is at most $2^{b-1}x^2(\log x)^{-b}$. This proves the bound. Integration by parts gives $I_b(x) = \frac{1}{2}x^2(\log x)^{-b} + \frac{b}{2}I_{b+1}(x) + O(1)$; applying this once more to I_{b+1} and the bound to I_{b+2} gives the expansion. $\square$

Proof of Theorem 1.4. By (9), $G(s) = F(s)\zeta(s)^{-1/2} = \prod_p E_p(s)$ with

$$E_p(s) = \left(1 - \frac{1 + p^{-1}}{2} u \right) (1 - u)^{-1/2}, \quad u = p^{-s},$$

where $(1 - u)^{-1/2}$ is the principal branch. For $\sigma \geq \frac{3}{4}$ we have $|u| \leq 2^{-3/4}$, and expanding $(1 - u)^{-1/2} = 1 + \frac{u}{2} + O(|u|^2)$ gives $E_p(s) = 1 - \frac{u}{2p} + O(|u|^2) = 1 + O(p^{-7/4} + p^{-3/2})$ uniformly. So the product converges absolutely and uniformly on $\sigma \geq \frac{3}{4}$, and G is holomorphic and bounded there. Lemma 5.1 applies with $z = \frac{1}{2}$, $\delta = 1$ and any c_0 for which $D \subseteq \{\sigma > \frac{3}{4}\}$, and gives

$$S(t) := \sum_{n \leq t} g(n) = \frac{t}{\sqrt{\log t}} \left(\lambda_0 + \frac{\lambda_1}{\log t} \right) + O\left(\frac{t}{(\log t)^{5/2}} \right) \quad (t \geq 3), \quad (10)$$

with $\lambda_0 = G(1)/\sqrt{\pi}$. At $s = 1$, $E_p(1) = (1 + \frac{1}{2p})(1 - \frac{1}{p})^{1/2}$, and a short computation gives

$$\left. \frac{d}{ds} \log E_p(s) \right|_{s=1} = \log p \left(\frac{p+1}{(2p+1)(p-1)} - \frac{1}{2(p-1)} \right) = \frac{\log p}{2(2p+1)(p-1)},$$

⁴We also differentiated $\log L(s, \chi_{-4})$ numerically at $s = 1$; the two values agree to ten decimal places.

so $G'(1)/G(1) = \frac{1}{2} \sum_p \log p / ((2p+1)(p-1))$.

By partial summation, $\sum_{n \leq x} \rho(n) = \sum_{n \leq x} n g(n) = xS(x) - \int_1^x S(t) dt$. On $[1, 3]$ we have $S(t) \leq t$, and on $[3, x]$ we insert (10) and use Lemma 5.2 with $b = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$:

$$\int_1^x S(t) dt = \frac{x^2}{2\sqrt{\log x}} \left(\lambda_0 + \frac{\lambda_0/4 + \lambda_1}{\log x} \right) + O\left(\frac{x^2}{(\log x)^{5/2}} \right).$$

Subtracting this from $xS(x)$ gives

$$\sum_{n \leq x} \rho(n) = \frac{x^2}{2\sqrt{\log x}} \left(\lambda_0 + \frac{\lambda_1 - \lambda_0/4}{\log x} \right) + O\left(\frac{x^2}{(\log x)^{5/2}} \right).$$

Thus $C_2 = \lambda_0/2 = G(1)/(2\sqrt{\pi})$ and, by (6), $\kappa_2 = \lambda_1/\lambda_0 - \frac{1}{4} = \frac{1}{4} - \frac{\gamma}{4} - \frac{1}{2} G'(1)/G(1)$, which are the stated formulas. $\square$

The Euler product $G(1) = \prod_p (1 + \frac{1}{2p})(1 - \frac{1}{p})^{1/2} = 0.812105711163122\dots$ converges slowly, so we evaluated it, and the sum in κ_2 , through the prime zeta function $\mathcal{P}(s) = \sum_p p^{-s}$:

$$\log G(1) = \sum_{k \geq 2} \frac{(-1)^{k+1} 2^{-k} - \frac{1}{2}}{k} \mathcal{P}(k), \quad \sum_p \frac{\log p}{(2p+1)(p-1)} = -\frac{1}{3} \sum_{k \geq 0} \left(1 - \left(-\frac{1}{2}\right)^{k+1} \right) \mathcal{P}'(k+2).$$

Our value of $G(1)$ agrees with all 25 digits given in [4].

5.3 Numerical check

We computed $\mathcal{E}(x)$ exactly for $x \leq 2 \cdot 10^{10}$ with a bit sieve that removes the odd multiples of primes $p \equiv 1 \pmod{4}$, and $\sum_{n \leq x} \rho(n)$ exactly for $x \leq 10^{10}$ with a segmented sieve for $\varphi(n)$ and $\omega(n)$. Since $2\rho(n)$ is always an integer, the second sum was accumulated in exact 128-bit integer arithmetic.⁵ Tables 2 and 3 and Figures 3 and 4 compare the counts with the one-term and two-term predictions

$$\mathcal{E}_1(x) = \frac{Cx}{\sqrt{\log x}}, \quad \mathcal{E}_2(x) = \mathcal{E}_1(x) \left(1 + \frac{\kappa}{\log x} \right), \quad R_1(x) = \frac{C_2 x^2}{\sqrt{\log x}}, \quad R_2(x) = R_1(x) \left(1 + \frac{\kappa_2}{\log x} \right).$$

The one-term ratio $\mathcal{E}(x)/\mathcal{E}_1(x)$ is still 1.032 at $x = 2 \cdot 10^{10}$, which is the slow $1/\log x$ convergence one expects. The two-term prediction removes most of the gap. If the expansion is carried one step further, $\mathcal{E}(x)/\mathcal{E}_2(x) - 1 = \kappa_3(\log x)^{-2} + O((\log x)^{-3})$, and the same computation as in the proof of Theorem 1.3, with $Z''(1)$ and $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$, gives $\kappa_3 = 1.2639\dots$. The last column of Table 2 is consistent with this: over all our checkpoints $10^6 \leq x \leq 2 \cdot 10^{10}$, including those not shown in the table, it lies between 1.46 and 1.87, and apart from small fluctuations it falls from 1.74 at $x = 10^6$ to 1.46 at $x = 2 \cdot 10^{10}$, towards κ_3 . For the mean value κ_2 is small, and the one-term prediction is already within 0.13% at $x = 10^{10}$. The corresponding third coefficient is 0.0727..., and over all checkpoints $10^6 \leq x \leq 10^{10}$ the last column of Table 3 lies between 0.106 and 0.148 and falls from 0.139 at $x = 10^6$ to 0.106 at $x = 10^{10}$.

6 Conclusions

The ratio of surjective group homomorphisms to ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ is $2^{\varepsilon(n)} |U_n^2|$, with $\varepsilon(n)$ determined by the 2-part of n , and it is an integer exactly off the set $\mathcal{E}$ of $n \equiv 2 \pmod{4}$ whose odd part has only prime factors $\equiv 3 \pmod{4}$. For odd n the ring homomorphisms are in

⁵Both programs are short C files; the larger run took about nine minutes on one core. They are included in the archive listed in the data availability statement.

Table 2. The exceptional set: exact counts against Theorem 1.3.

x	$\mathcal{E}(x)$	$\mathcal{E}_1(x)$	$\mathcal{E}_2(x)$	$\mathcal{E}/\mathcal{E}_1$	$\mathcal{E}/\mathcal{E}_2$	$(\mathcal{E}/\mathcal{E}_2 - 1)(\log x)^2$
10^4	886	802	863	1.10535	1.02679	2.27
10^5	7729	7169	7608	1.07807	1.01588	2.10
10^6	69413	65447	68785	1.06061	1.00913	1.74
10^7	636229	605918	632411	1.05003	1.00604	1.57
10^8	5911404	5667842	5884687	1.04297	1.00454	1.54
10^9	55450285	53436931	55254205	1.03768	1.00355	1.52
10^{10}	523916546	506947235	522463396	1.03347	1.00278	1.47
$2 \cdot 10^{10}$	1031330156	998969959	1028651968	1.03239	1.00260	1.46

Table 3. The mean value of ρ : exact sums against Theorem 1.4.

x	$\sum_{n \leq x} \rho(n)$	$\sum \rho/R_1$	$\sum \rho/R_2$	$(\sum \rho/R_2 - 1)(\log x)^2$
10^4	$7.587531 \cdot 10^6$	1.005150	1.002447	0.208
10^5	$6.775787 \cdot 10^8$	1.003564	1.001404	0.186
10^6	$6.179019 \cdot 10^{10}$	1.002525	1.000727	0.139
10^7	$5.717876 \cdot 10^{12}$	1.002037	1.000496	0.129
10^8	$5.346803 \cdot 10^{14}$	1.001704	1.000356	0.121
10^9	$5.039785 \cdot 10^{16}$	1.001460	1.000261	0.112
10^{10}	$4.780294 \cdot 10^{18}$	1.001278	1.000200	0.106

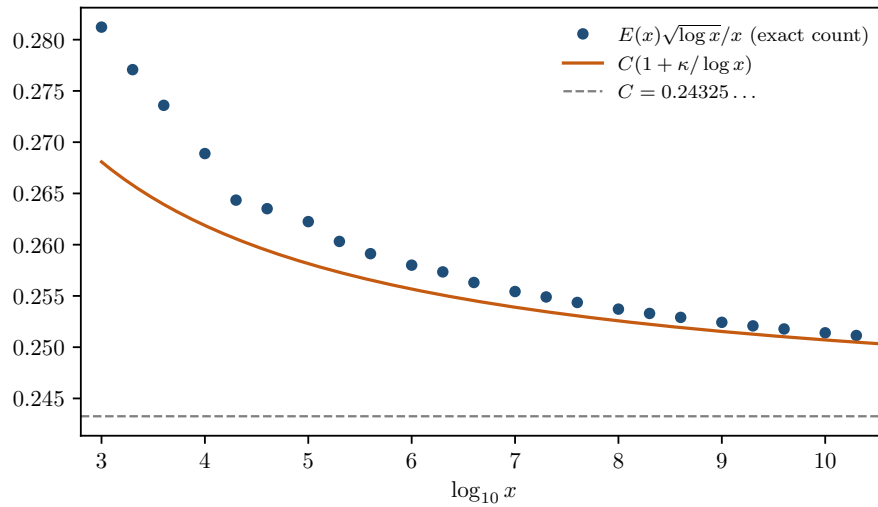


Figure 3. $\mathcal{E}(x)\sqrt{\log x}/x$ for $10^3 \leq x \leq 2 \cdot 10^{10}$, $x = c \cdot 10^k$ with $c \in \{1, 2, 4\}$, against the limit C and the two-term curve $C(1 + \kappa/\log x)$ of Theorem 1.3.

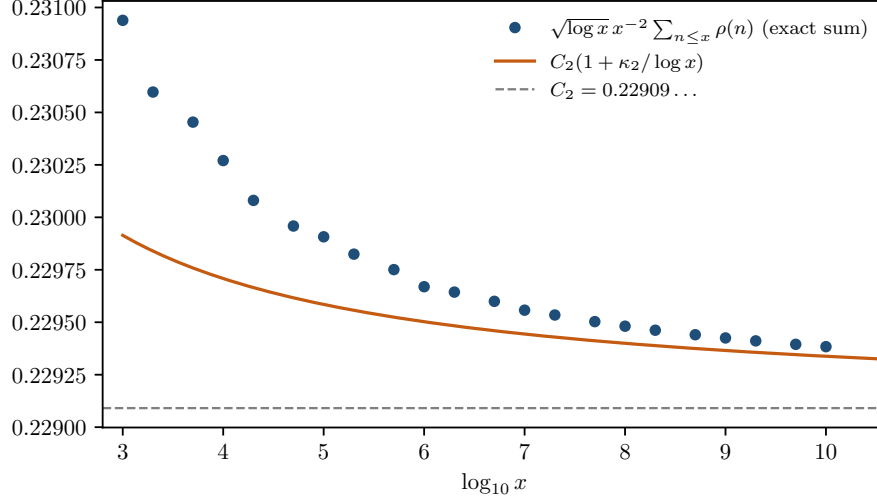


Figure 4. $x^{-2} \sqrt{\log x} \sum_{n \leq x} \rho(n)$ for $10^3 \leq x \leq 10^{10}$, $x = c \cdot 10^k$ with $c \in \{1, 2, 5\}$, against the limit C_2 and the two-term curve $C_2(1 + \kappa_2/\log x)$ of Theorem 1.4.

canonical bijection with the subfields of $\mathbb{Q}(\zeta_n)$ of degree at most 2, and $\rho(n)$ is the degree of $\mathbb{Q}(\zeta_n)$ over its maximal multiquadratic subfield. The set $\mathcal{E}$ has $Cx(\log x)^{-1/2}(1 + \kappa/\log x + O((\log x)^{-2}))$ elements up to x , and the partial sums of ρ are $C_2 x^2 (\log x)^{-1/2}(1 + \kappa_2/\log x + O((\log x)^{-2}))$, with all four constants explicit and confirmed by exact computation to $2 \cdot 10^{10}$ and 10^{10} respectively. The same method gives further terms of both expansions, at the cost of higher derivatives of $L(s, \chi_{-4})$ and of the Euler products at $s = 1$.

Author contributions

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Use of AI tools declaration

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Conflict of interest

The authors declare no conflicts of interest.

Data availability

The C and Python programs and the exact counts behind Tables 2 and 3 are available at <https://github.com/creelie/quadratic-residue>.

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